

Lecture 3: Measures of Central Tendency

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Table of contents

- 1 Learning Outcomes
- 2 Arithmetic Mean
 - Example
- 3 Median
 - Two-step process to find the median
 - Example
- 4 Mode
- 5 Central value for Grouped data
 - Median for grouped data
 - Mode for grouped data
- 6 Weighted Mean (WM)
 - Example
- 7 Group work Homework

Learning Outcomes

- Compute the different types of mean
 - i. Arithmetic mean
 - ① Ungrouped data
 - ② Group data
 - ii. Geometric mean
 - iii. Harmonic mean
 - iv. Weighted mean
 - v. Trimmed mean
- Compute the median and mode of a set of data.
 - ① Ungrouped data
 - ② Group data
- Properties of a good Measure of Central tendency

Arithmetic Mean

- The Arithmetic Mean of a set of n numbers.

$$AM = \frac{x_1 + \dots + x_n}{n}$$

- Arithmetic Mean for population and sample

$$\mu = \frac{\sum_{i=1}^N (x_i)}{N}$$

$$\bar{x} = \frac{\sum_{i=1}^n (x_i)}{n}$$

An example

- ◇ **Example:** A sample of five executives received the following bonuses last year (\$000) : 14.0, 15.0, 17.0, 16.0, 15.0
- ◇ **Problem:** Determine the average bonus given last year.
- ◇ **Solution:**

$$\bar{x} = \frac{14.0 + 15.0 + 17.0 + 16.0 + 15.0}{5}$$

$$\bar{x} = \frac{77}{5} = 15.4$$

Properties of Arithmetic Mean

- Easy to compute
- It requires at least the interval scale
- All values are used
- It is unique
- The sum of the deviations from the mean is 0
- It is easily affected by extremes, such as very big or small numbers in the set (non-robust).

Median

- The Median is the midpoint of the values after they have been ordered from the smallest to the largest
- Equivalently, the Median is a number which divides the data set into two equal parts, each item in one part is no more than this number, and each item in another part is no less than this number.

Two-step process to find the median

Step 1. Sort the data in a nondecreasing order

Step 2.

- ◇ If the total number of items n is an odd number, then the number on the $(n + 1)/2$ position is the median;
- ◇ If n is an even number, then the average of the two numbers on the $n/2$ and $n/2 + 1$ positions is the median. (For ordinal level of data, choose any one on the two middle positions).

Example

- **Example 1:** The ages for a sample of five college students are: 21, 25, **19**, 20, 22
- Arranging the data in ascending order gives:
19, 20, 21, 22, 25.
- The median is 21.
- **Example 2:** The heights of four basketball players, in inches, are: 76, 73, 80, 75
- Arranging the data in ascending order gives:
73, **75, 76**, 80.

The median is the average of the two middle numbers

$$\text{Median} = \frac{75 + 76}{2} = 75.5$$

Properties of Median

- It requires at least the ordinal scale
- All values are used
- It is unique
- It is easy to calculate but does not allow easy mathematical treatment
- It is not affected by extremely large or small numbers (robust)

Mode

- The number that has the highest frequency.
- **Example:** The exam scores for ten students are:
81, 93, 84, 75, 68, 87, 81, 75, 81, 87
- The score of 81 occurs the most often. It is the Mode.

Properties of Mode

- Even nominal data have mode(s)
- All values are used
- It is not unique
 - **Modeless**: if all data have different values, such as 1, 1, 1
 - **Multimodal**: if more than one value have the same frequency, such as 1, 1, 2, 2, 3.
- It is easy to calculate but does not allow easy mathematical treatment
- It is not affected by extremely large or small numbers (robust)

Mean for grouped data

- The mean of a sample of data organized in a frequency distribution is computed by the following formula:

$$\bar{x} = \frac{f_1 x_1 + \dots + f_k x_k}{f_1 + \dots + f_n}$$

- where f_i is the frequency of Class i and x_i is the class mid-point of Class i .

Example

- ① **Example:** Recall the weight example from Chapter 2:

Weight	F	mid point (x_i)	$f_i x_i$
130 - 136	2	133	266
137 - 143	5	140	700
144 - 150	9	147	1323
151 - 157	11	154	1694
158 - 164	6	161	966
165 - 171	5	168	840
172 - 178	2	175	350
	40		6139

- ② The mean for the grouped data is:

$$\bar{x} = \frac{6139}{40} = 153.475$$

Median for grouped data: Two-step procedure

Step 1. Identify the median class, which is the class that contains the number on the $n/2$ position.

Step 2. Estimate the median value within the median class using the following formula:

$$\text{median} = L + \frac{\frac{n}{2} - CF}{f} \times i$$

where

- ◇ L is the lower class boundary of the median class
- ◇ CF is the cumulative frequency before the median class
- ◇ f is the frequency of the median class
- ◇ i is the class interval or size

The Median for Grouped Data

- ① **Example:** Recall the weight example from Chapter 2:

Weight	F	CF
130 - 136	2	2
137 - 143	5	7
144 - 150	9	16
151 - 157	11	27
158 - 164	6	33
165 - 171	5	38
172 - 178	2	40

- ② The median for the grouped data is:

$$\text{median} = 150.5 + \frac{20 - 16}{11} \times 7 \approx 153.045$$

Mode for grouped data: Two-step procedure

Step 1. Identify the modal class, which is the class(es) that has the highest frequency(ies).

Step 2. Estimate the mode(s) within the modal class (es) using the following formula:

$$mode = L + \frac{f_1 - f_0}{2 \times f_1 - f_0 - f_2} \times i$$

where

- ◇ L is the lower class boundary of the modal class
- ◇ f_0 is the frequency above the modal class
- ◇ f_1 is the frequency of the modal class
- ◇ f_2 is the frequency below the modal class
- ◇ i is the class interval or width

The Mode for Grouped Data

- ① **Example:** Recall the weight example from Chapter 2:

Weight	Frequency
130 - 136	2
137 - 143	5
144 - 150	9
151 - 157	11
158 - 164	6
165 - 171	5
172 - 178	2

- ② The median for the grouped data is:

$$mode = 150.5 + \frac{11 - 9}{2 \times 11 - 9 - 6} \times 7 \approx 149.92$$

Weighted Mean (WM)

- The Weighted Mean (WM) of a set of n numbers

$$WM = \frac{w_1x_1 + \dots + w_nx_n}{w_1 + \dots + w_n}$$

- Example:** During an one hour period on a hot Saturday afternoon chris served fifty drinks. He sold:

No. of drinks	Price (Ksh)
5	50.00
15	75.00
15	90.00
15	110.00

Example

- **Problem::** compute the weighted mean of the price of the drinks
- The Weighted Mean (WM) of a set of n numbers

$$WM = \frac{5(50) + 15(75) + 15(90) + 15(110)}{5 + 15 + 15 + 15}$$



$$\bar{x} = \frac{4375}{50} = Ksh.87.5$$

Homework

- **By using a suitable example, discuss the properties, uses, advantages and limitations of the following;**
 - ◇ Geometric mean
 - ◇ Harmonic mean
 - ◇ Trimmed mean
- **Due date : Next class**